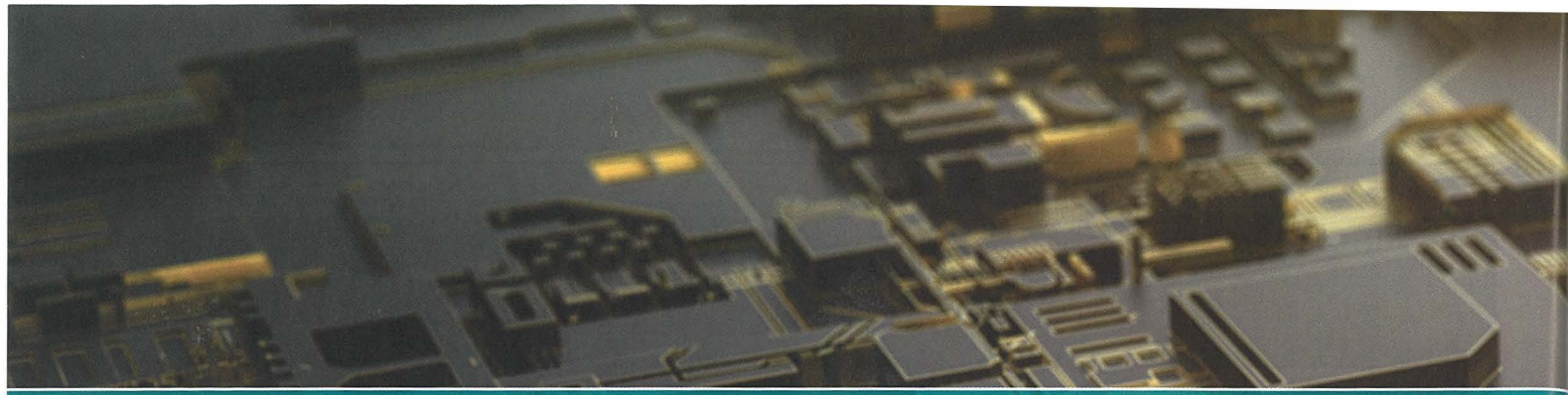


Chapter 3
Hardware



IN THIS CHAPTER YOU WILL:

- develop an understanding of the role of the central processing unit (CPU) in a computer, including the fetch-decode-execute cycle
- learn the role of each of the components in a CPU that has a Von Neumann architecture
- develop an understanding of what an embedded system is and be able to identify devices in which they are used
- understand what an input device and an output device is and why they are needed
- learn the role of a sensor and the purpose of a range of different sensors, including how they capture data
- learn the role of random access memory (RAM) and read only memory (ROM)
- learn the role of secondary storage including the operation of the different types
- develop an understanding of virtual memory and how it is created
- learn how cloud storage is used and the advantages and disadvantages of storing data in the cloud
- study the main components and types of address that are used when connecting a computer to a network.

GETTING STARTED

Describe to a partner how you think a computer processes an instruction. Ask your partner how they think a computer processes an instruction. Do you both agree? If not, discuss the differences you have and come to an agreed process and write this down. You can check at the end of the chapter how accurate your initial understanding was.

THE INTERNET OF THINGS

The use of computer hardware has become a major part of our daily lives. Have a quick think about how many pieces of computer hardware that you use in a day. You might find it's as many as 10 or more pieces of hardware! It is now possible to get many devices for our home that have computer hardware built into them. Can you think of any in your house?

We often refer to household devices that have computer hardware built into them as smart devices. It is possible to buy a smart refrigerator. This has features such as scanning and keeping a list of all food in the refrigerator. You can quickly and simply view what is available and when it might go out of date. Think about how that could reduce food waste! Smart refrigerators are also capable of keeping a list of items that are regularly used and can even keep a groceries list and use the internet to order items as they are required. They can also provide you with recipe ideas for the ingredients that you have. How awesome is that! We have to think about so many different things in our lives, so having assistance that can keep track of simple needs like this, people often find very helpful.

When people use smart devices, such as a smart refrigerator, it is often possible to connect them with other devices they own, such as their mobile telephone. This way they could get a list of what is in their refrigerator, when away from home, and can easily find out what they need to buy from the shop. The connecting of hardware devices in this way is called the **Internet of Things**.

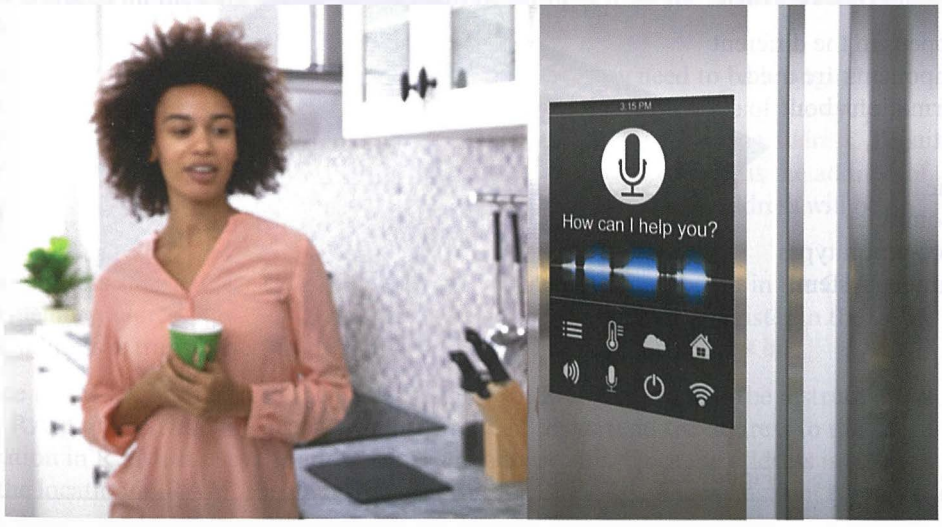


Figure 3.1: Smart refrigerators are part of the Internet of Things

KEY WORD

Internet of Things: the connection of computer systems and everyday devices, using the internet, to allow data to be exchanged.

CONTINUED

Discussion questions

1 What kind of feature would you include in a smart refrigerator that you would find helpful?

2 Can you see any problems that might arise as a result of connecting devices and the development of the Internet of Things?

3 Work with a partner and design your own smart device. What would you want it to do? What computer hardware would it need?

KEY WORDS

central processing unit (CPU): a component in a computer system that processes data and instructions.

microprocessor: an integrated circuit that is able to perform many of the functions of a CPU.

embedded system: a computer system that performs a dedicated function.

fetch-decode-execute cycle: the cycle through which data and instructions are processed.

architecture: the design of a computer system, including the components it contains.

3.1 The role of the CPU

The **CPU (central processing unit)** is one of the fundamental parts of a computer. Many people say it is the most important part, as it is where all the data and instructions are processed. This means it is a little bit like your brain, as that is where you process all the data and instructions that you sense on a daily basis.

All computers have a form of CPU. In devices such as personal computers (PC) and laptops, this is normally a separate component in the computer. In devices such as those found in our houses, e.g. our washing machine, a similar type of component exists called a **microprocessor**. This performs a similar role to a CPU, but on a smaller scale. Whereas a CPU can process thousands of complex instructions, a microprocessor will process a much more limited set of instructions. A general purpose computer, such as a PC or a laptop, has a CPU and is able to perform a wide variety of tasks. A microprocessor is often built into a system that is designed to perform a very specific task. This type of system is called an **embedded system**. There are examples of embedded systems in many different areas of our lives, for example, a traffic light, a washing machine and a digital alarm clock.

When an instruction is processed by a computer, the method that is used to do this is called the **fetch-decode-execute cycle**.

You need to understand what happens in the different stages of the cycle and which components are used. You'll learn some interesting information about how a computer works. How exciting!

You will learn the operation of the **architecture** of a Von Neumann computer system, which is one of the most common types of modern architecture for a computer system.

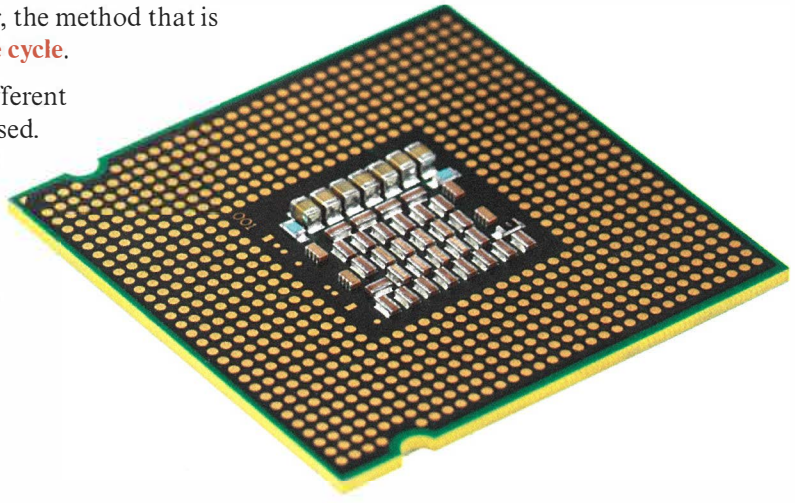


Figure 3.2: A CPU chip

ACTIVITY 3.1

Use the internet to research who Von Neumann was and why he developed the Von Neumann architecture.

This is a basic representation of the fetch-decode-execute cycle:

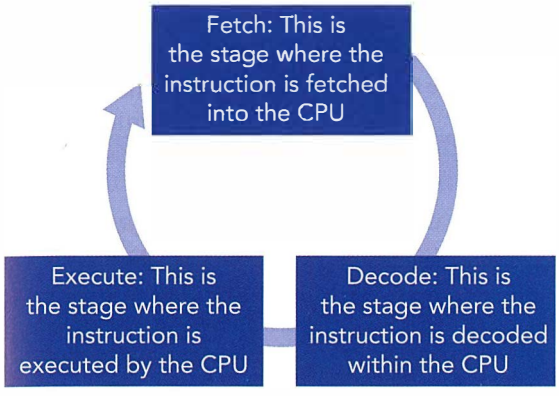


Figure 3.3: The fetch-decode-execute cycle

Individual processes happen at each of these stages, using different components, to carry out the cycle. So, what happens at each stage? We will explore this in the next sections.

3.2 The fetch stage

Data and instructions are given to the computer through the use of **input devices**. You will learn about these later in the chapter. When the data and instructions are input, they are stored in **RAM (random access memory)**. If a user wants to open a file that is stored on the **hard drive**, this data will be brought from the hard drive into the RAM, also.

In order for the data and instructions to be processed, they need to be brought from the RAM into the CPU. This is what happens at the fetch stage of the cycle. A component within the CPU, called the **program counter (PC)**, stores an **address**. This component is an example of a register. The address stored by the PC is the address of the next instruction that needs to be processed. The details of this address will be a location within the RAM.

This address stored in the PC needs to be sent to another component in the CPU called the **memory address register (MAR)**. This is another example of a register in the CPU. The address is transported from the PC to the MAR using the **address bus**.

Once the MAR has received the address it now knows where to find the instruction in the RAM. To retrieve the instruction the MAR needs to send the address to the location in RAM. It uses the address bus again to do this. Once the address is received at the location in RAM, the data stored in this location, e.g. the instruction, is sent to a

KEY WORDS

input device: a device that allows data to be entered into a computer system.

RAM (random access memory): a component in the CPU that holds data and programs that are currently in use.

hard drive: a storage device that stores data permanently, until it is deleted by the user.

program counter (PC): a component in the CPU that stores the address of the next instruction to be processed.

address: a memory location in RAM.

memory address register (MAR): a component in the CPU that holds the address of the data or instruction to be located in RAM.

address bus: wires used for the transmission of addresses between components in a computer.

component within the CPU called the **memory data register (MDR)**. This is also an example of a register in the CPU. The instruction is sent to the MDR using the **data bus**. When the MDR has received the instruction, it sends it to the **current instruction register (CIR)**. This is another example of a register in the CPU. The MDR uses the data bus to send the instruction to the CIR. This register is part of a component within the CPU called the **control unit (CU)**. It is the CU that is responsible for the next stage of the cycle, the decode stage.

Figure 3.4 shows the operation of the fetch stage of the cycle:

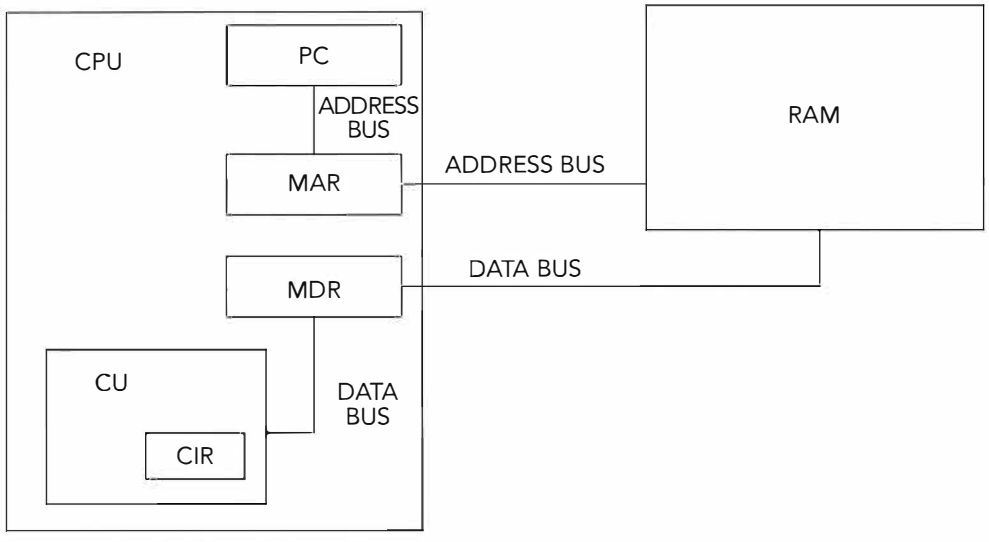


Figure 3.4: How the fetch stage operates

KEY WORDS

memory data register (MDR): a component in the CPU that holds the data or instruction that are fetched from RAM.

data bus: wires used for the transmission of data and instructions between components in a computer.

current instruction register (CIR): a register that is built into the CU that holds the current instruction that is being processed in the CPU.

control unit (CU): the component in the CPU that controls all the operations in the CPU.

instruction set: a set of commands that can be processed by a certain CPU.

ACTIVITY 3.2

Draw a diagram like the one given in Figure 3.4 to see if you can draw and label the process for the fetch stage of the cycle.

3.3 The decode stage

The decode stage of the cycle begins when the instruction is received by the CU. The CU is the unit in the CPU that decodes the instruction. The instruction needs to be decoded so that the CPU can understand what is required to execute the instruction. To decode an instruction the CU uses an **instruction set**. This is a set of all the commands that are understood and can be processed by the CPU. These commands are in machine code. Once the instruction is decoded using the instruction set, it can now be executed.

3.4 The execute stage

The execute stage is when any actions that are required for the instruction are carried out. This is likely to involve some mathematical and logical calculations. If these kinds

of calculations are required, the data used in the process is sent to the **arithmetic logic unit (ALU)**. This is a component within the CPU that is responsible for any calculations. Interim values may be created whilst these calculations are being performed, the ALU has a special register built into it called the **accumulator (ACC)**. It is this special register that will store the interim values that are created in any calculations. Once any calculations have been performed, the actions for the instruction can be performed, e.g. the file can be opened or the letter typed can be displayed.

How do all the different components know what they are doing in the fetch-decode-execute cycle? They have a conductor! This conductor is the CU. It sends signals to all the components within the CPU to tell them when they need to perform their different roles. The CU uses the **control bus** to send these signals.

KEY WORDS

arithmetic logic unit (ALU): the component in the CPU that performs all the mathematical and logical operations required when processing data and instructions.

accumulator (ACC): a register that is built into the ALU that stores the result of any interim calculations.

control bus: wires used for the transmission of control signals between components in a computer.

ACTIVITY 3.3

Add the ALU and the ACC to your diagram (from Activity 3.2) then label it to show the process for the decode and execute stages of the cycle.

Peer Assessment

Exchange diagrams from Activities 3.2 and 3.3 with a friend and use the description of the fetch-execute-decode cycle to check whether their diagram is correct.

Questions

1 Where is data stored before it is fetched into the CPU?

2 Which component in the CPU is responsible for making sure the fetch-decode-execute cycle is performed correctly?

3 What is the role of a bus?

4 Why does the ALU need the ACC? What would have to happen if it didn't have this special register?

5 Copy the table below and fill in a description for each component in the CPU:

Component	Description
Program counter (PC)	
Memory address register (MAR)	
Memory data register (MDR)	
Control unit (CU)	
Current instruction register (CIR)	
Arithmetic logic unit (ALU)	
Accumulator (ACC)	
Address bus	
Data bus	
Control bus	

3.5 Factors that can affect the performance of a CPU

The performance of a CPU refers to how quickly the CPU can process an instruction. This is important as the more instructions that can be processed in a second, the higher the performance of the CPU. Users often want a high performing CPU, so they are not sat waiting and frustrated with their computer because it takes too long to open a document, for example.

There are three main factors that can affect the performance of a CPU and this is the number of **cores** it contains, the **clock speed** and the size of the **cache**.

A single core in a CPU contains all the components that you have learnt that are used to perform the fetch-decode-execute cycle. Therefore, if a CPU has one core, it can perform one fetch-decode-execute cycle at a time. If a CPU has two cores (this is often called a dual core processor) it can perform two fetch-decode-execute cycles at the same time, one on each core. Some CPUs have four cores (this is often called a quad core processor) and can perform four fetch-decode-execute cycles all at the same time. Imagine completing a task: if you can only do one thing at a time to complete the task, it might take you quite a long time but if you could do four things at once to complete the task, you will complete it much quicker. Therefore, if a CPU has multiple cores, its performance can be increased.

Multiple cores also allow a computer to multitask. If it can process more than one instruction at a time, it can perform more than one task at a time.

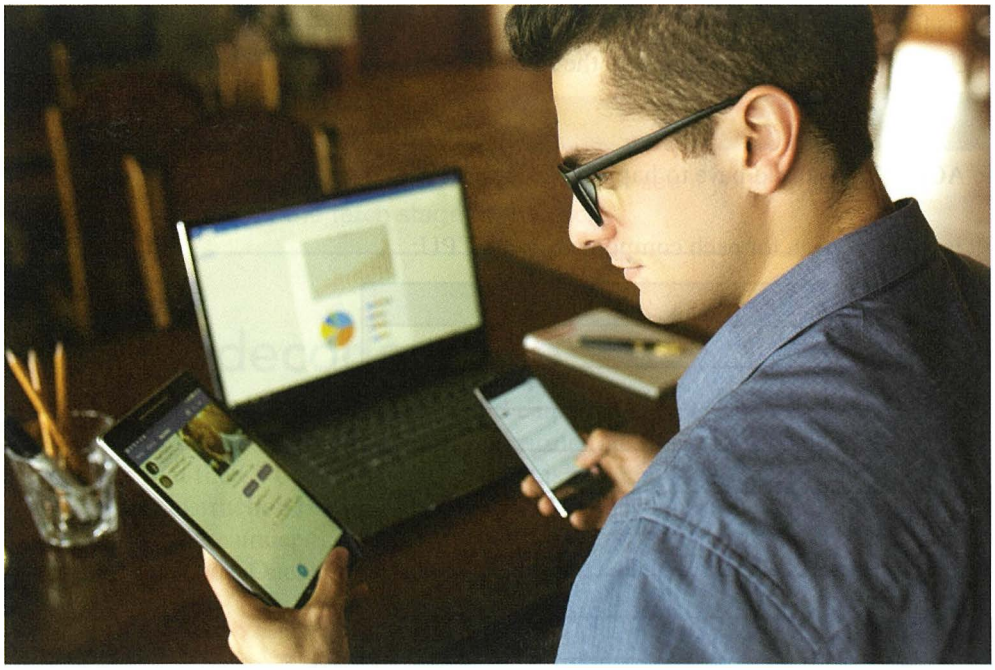


Figure 3.5: Multiple cores allow a computer to multitask

KEY WORDS

core: the part of the CPU that contains all the components that are used to perform the fetch-decode-execute cycle.

clock speed: the number of fetch-decode-execute cycles that can be performed in a second.

cache: a type of storage that is built into the CPU, to store the most frequently used data and instructions.

ACTIVITY 3.4

The performance of a CPU is not always increased by the addition of more cores. Use the internet to find out why the performance may not always be increased.

A CPU contains an internal clock that controls the speed at which instructions are processed. The speed is measured in hertz. A CPU that has a clock speed of 1 hertz is able to process one instruction per second. If your CPU had a clock speed of 1 hertz you'd be waiting rather a long time for any tasks to be completed!

Most modern computers have a clock speed that is measured in gigahertz (GHz). A CPU with a clock speed of 2 GHz can process 2 billion instructions a second. That is definitely a busy CPU! If a computer has two cores, each core can process 2 billion instructions a second.

ACTIVITY 3.5

It is possible to change the speed of a CPU using overclocking. Use the internet to find out what is meant by overclocking.

A CPU also contains a cache. The cache is a type of data storage within the CPU. Instructions and data that the computer uses most can be stored in the cache. This means that the CPU doesn't need to fetch this data from the CPU, just from the cache, so that process will be quicker. If the CPU can get the data and instructions quicker, the performance of processing those instructions will improve. If a computer has a larger cache, so it has more storage space to store the instructions and data it uses most, then it can store more of them, so will need to spend less time going to the RAM.

COMPUTER SCIENCE IN CONTEXT

Some people need very powerful processors in their computer, as they need to process a lot of data in a very short time period. One example would be an architect. They need to be able to use a computer to create a 3D model of their building design. It takes a lot of data to render these models, so a very powerful processor is needed to do this. 'Render' simply means to make the model. The computer does this by putting it together bit by bit.

Questions

6 Which CPU can process more instructions at the same time, a quad core or a dual core?

7 How many instructions per second can a dual core 2.4 Ghz CPU process?

8 If the cache size of a CPU is too big, this can actually have a negative effect on the performance of the CPU. Why might this happen?

3.6 Input and output devices

You now know how a computer processes data, but how does it receive that data and how do we see the result of the processing? This is possible through the use of input and **output devices**.

An **input device** is any device that allows data to be entered into a computer system. This data could take many forms including text, images and sound.

Common examples of input devices are:

- barcode scanner
- digital camera
- keyboard
- microphone
- optical mouse
- QR code scanner
- scanner (2D and 3D)
- touch screen (resistive, capacitive and infra-red).

KEY WORDS

output device: a device that can be used to obtain the results of data that has been processed.

input device: a device that allows data to be entered into a computer system.

Question

9 Copy the table below that includes the examples of the input devices. For each input device, describe the kind of data that it allows to be input and give an example of when it is used. The first one has been done for you.

Input device	Description of data input	Example of use
Barcode scanner	*It scans a barcode so that the data stored in the barcode can be obtained.*	*It is used in a supermarket to get the price of a product and as part of a stock control system.*
Digital camera		
Keyboard		
Microphone		
Optical mouse		
QR code scanner		
Scanner (2D and 3D)		
Touch screen (resistive, capacitive and infra-red)		

An output device is any device that allows the result of the data processing to be seen or heard.

Common examples of output devices are:

- actuator
- digital light processing (DLP) projector

- inkjet printer
- laser printer
- light emitting diode (LED) screen
- liquid crystal display (LCD) projector
- liquid crystal display (LCD) screen
- speaker
- 3D printer.

Question

10 Copy the table below that includes the examples of the output devices. For each output device, describe the kind of data that it allows to be output and give an example of when it is used. The first one has been done for you.

Output device	Description of data output	Example of use
Actuator	*It is a component that outputs an action, often a type of movement, that causes another device to operate.*	*Can be used in an automated system to move or turn on/off another device, e.g. a light.*
Digital light processing (DLP) projector		
Inkjet printer		
Laser printer		
Light emitting diode (LED) screen		
Liquid crystal display (LCD) projector		
Liquid crystal display (LCD) screen		
Speaker		
3D printer		

Many computer systems that you encounter in your daily life are automated systems. You can learn more about what an automated system is and how they work in Chapter 6.

A type of input device called a **sensor** is used in an **automated system**. The sensor is used to capture different types of data to be processed by the system. The sensor captures analogue data from its immediate surrounding environment. This data can then be processed and analysed by a computer system. Sensors are normally set to capture data at given time intervals. This could be as constant as every single second or even fraction of a second. It could also be set to be every minute or at particular times of the day.

Table 3.1 shows the different types of sensor that you need to know, a description of the type of data that they capture and an example of a how they might be used.

KEY WORDS

sensor: a type of input device that is used to capture data from its immediate environment.

automated system: a system that is designed to operate without the need for any human intervention.

Type of sensor	Type of data captured	Example of its use
Acoustic	This type of sensors measures the level of sound in an environment.	These sensors are used in many applications that involve sound. An acoustic sensor can be used in a security system. It can be placed near a window and constantly measure the level of sound. If it captures a reading that shows a sudden increase in sound, this could mean that the window has been broken and the building may be at risk.
Accelerometer	This type of sensor measures acceleration forces. These may be static forces, such as the continual force of gravity. They can also be dynamic forces, such as those created by movement and vibrations.	These sensors are used in a wide variety of devices. Mobile phones use an accelerometer to know which way up it is faced, to automatically turn the screen on and off. They can be used to monitor for earthquakes, as they can capture the initial vibrations created. They can also be used in cars to sense when a crash has occurred, so that airbags can be inflated.
Flow	This type of sensor measures the amount of liquid, gas or steam that is flowing through or around a certain environment.	These sensors are often used in factories and sites such as nuclear power plants. They make sure that the liquid, gas or steam flows at a constant temperature through an environment, such as a pipe. This makes sure that the pipes don't rupture and break due to too much flowing through them.
Gas	This type of sensor measures the presence and concentration of a gas within the immediate atmosphere.	These sensors can be used in people's homes. They can be set to measure a certain gas, such as carbon monoxide. They constantly capture the data in the immediate atmosphere to see if too much carbon monoxide is present, which could endanger the health of anyone living in the home.
Humidity	This type of sensor measures the level of moisture in the immediate atmosphere.	These sensors can be used in farming and agriculture to make sure that the air in areas such as greenhouses has the correct level of moisture to provide the best growing conditions for the fruits and vegetables. They can also be used in places such as art galleries, to make sure that the humidity level is constant. Too much or too little can ruin paintings.
Infra-red	This type of sensor measures infra-red radiation. This type of radiation can be emitted in different amounts by both objects and people.	These sensors can also be used in security systems. This can be done in two different ways. The device containing the sensor can emit infra-red radiation and when this bounces back to the device, the readings can show from the distance it has travelled whether an intruder is present. It can also operate by capturing the infra-red radiation emitted by the intruder.
Level	This type of sensor measures whether a substance, such as a liquid, is at a certain level or amount.	These sensors can be used in a car to make sure that essential liquids, such as oil and fuel, do not get too low.
Light	This type of sensor measures the ambient light in a certain environment. It can also measure the presence of a direct light, such as a laser beam.	These sensors can be used in automatic lighting systems. Streetlights can be fitted with a light sensor that will allow the light to turn on in the evening, when it becomes darker.

(continued)

Type of sensor	Type of data captured	Example of its use
Magnetic field	This type of sensor measures the presence of magnetic field that may be emitted by an object.	These sensors can be used to count how many cars pass through a certain area, for example into a car park. The car will disrupt the Earth's naturally magnetic field as it passes over the sensor and the data can be captured by the sensor.
Moisture	This type of sensor measures the amount of water that is present in a substance, such as soil.	These sensors can also be used by farming and agriculture, to make sure that the fruits and vegetables have the best level of water in the soil to help them grow.
pH	This type of sensor measures the pH level of a substance.	These sensors can be used by environmental agencies to make sure that local lakes and river are not being polluted.
Pressure	This type of sensor measures the force of pressure that is applied to the sensor or device. This could be the pressure created by a solid object, or it could be created by liquid or gas.	These sensors could also be used in a security system. They can be placed at the base or sides of an opening, such as a window or a door. When that window or door is opened, the pressure will decrease, and the system will recognise that an intruder has entered.
Proximity	This type of sensor measures how close an object is in comparison to the sensor. It does this by emitting electromagnetic radiation or an electromagnetic field and measures the radiation as it returns to see if there are any changes.	These types of sensors can be used in robots in manufacturing. They allow the robots to measure how close they are to different objects, when moving around a factory.
Temperature	This type of sensor measures the temperature of an object or substance by either directly touching it or capturing data from the surrounding environment.	These types of sensors are used in air conditioning systems. They allow the temperature of a room to be kept at a certain level.

Table 3.1: Different types of sensor



Figure 3.6: An accelerometer can sense a crash so that airbags inflate

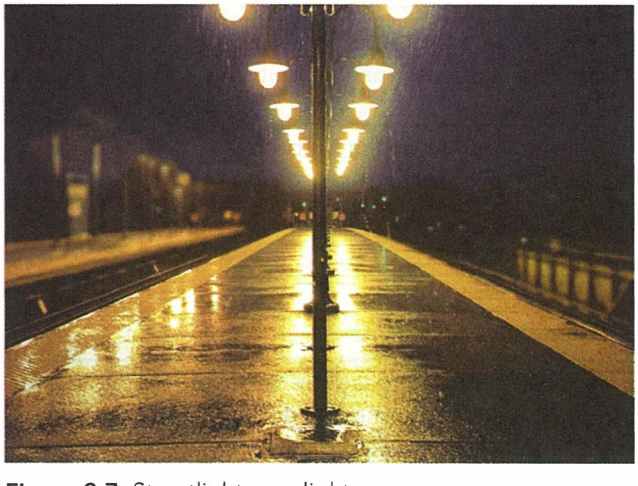


Figure 3.7: Streetlights use light sensors

Questions

11 Is a sensor an example of an input device or an output device?

12 Can you name **five** different types of sensor?

13 Can you think of one application for each of the following sensors?

- a Light
- b Magnetic field
- c Temperature

14 Can you think of **two** ways that the use of sensors is of benefit to humans?

15 How could an accelerometer be used to protect the hard drive on a computer?



Figure 3.8: Art galleries use temperature and humidity sensors

3.7 Data storage

You need to store data in a computer system on a constant basis. This might be temporarily, whilst it is being processed, or on a more permanent basis for future use. For these purposes, there are two main categories of storage that exist in a computer, they are **primary storage** and **secondary storage**.

Primary storage is a category of storage that is directly accessed by the CPU. This means that the CPU has a direct connection to it and that data is sent between the CPU and the primary storage using this connection. There are two main types of primary storage, random access memory (RAM) and **read only memory (ROM)**. There are several characteristics of each type of primary storage, these include those shown in Table 3.2:

Characteristics of RAM	Characteristics of ROM
It is volatile storage. This mean that it requires power to store data. If the power is turned off, the contents of RAM are lost. Volatile storage is sometimes referred to as temporary storage.	It is non-volatile. This means that if power is removed, the contents of the ROM are retained. Non-volatile storage is sometimes referred to as permanent storage.
It stores data and programs that are currently in use by the computer.	It stores programs that initially boot the computer, such as the bootstrap and the BIOS.
Its contents are constantly being changed as data is processed by the CPU.	The contents of ROM are fixed and do not normally change. It is possible to reprogram it, but this is a complex process and only normally attempted in circumstance when a computer system is failing to boot correctly.
It is possible to increase the amount of RAM a computer has by adding further RAM components	It is not easy to alter the ROM available in a computer system.

Table 3.2: Characteristics of primary storage

KEY WORDS

primary storage: a type of storage that is directly accessed by the CPU.

secondary storage: a type of storage that is designed to store data permanently, until it is deleted by the user.

ROM (read only memory): a type of primary storage that stores the start up instruction for the computer.

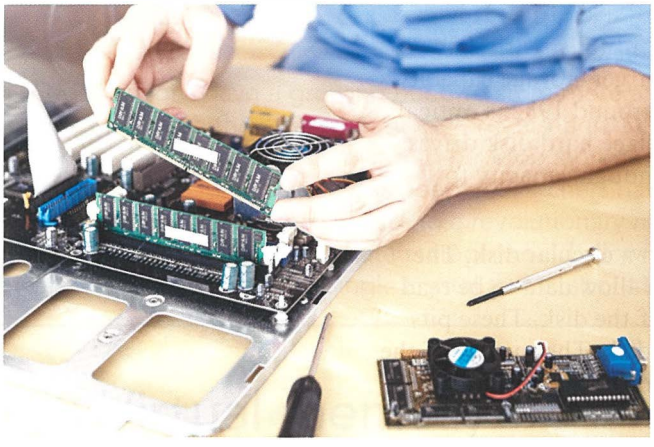


Figure 3.9: Technician changing a computer's RAM



Figure 3.10: The inside of a hard disk drive

ACTIVITY 3.6

Did you know that there are also different types of RAM and ROM? You do not need to know what these are for the exam, but it is interesting to find out what they are. Use the internet to research the different types of RAM and ROM and when they are used.

Secondary storage is a category of storage that is not directly accessed by the CPU. This means that data and instructions need to be sent from secondary storage into primary storage in order for it to be processed. Secondary storage is also a type of non-volatile storage. Can you remember what this means? There are three main types of secondary storage, these are **magnetic**, **optical** and **solid-state storage**.

Magnetic storage relies on the properties of magnetism to store data. A modern magnetic storage device normally has circular plates called platters that spin to allow data to be read and written. The platters are separated into tracks and sectors. It has a read/write head that moves across the platters to read or write the data. It uses an electromagnet to magnetise dots on each platter that store the data. If a dot is magnetised it stores the binary value 1. If a dot is demagnetised it stores the binary value 0. Each platter has billions of these magnetised dots. An example of this type device is called a hard disk drive.

ACTIVITY 3.7

You have just learnt that a hard disk drive is a type of magnetic storage and how it operates. There are older types of magnetic storage called magnetic tape and floppy disks. What can you find out about them?

Can you also find out what the largest storage capacity is for a hard disk drive?

KEY WORDS

magnetic storage: a type of secondary storage that uses the properties of magnetism to store data.

optical storage: a type of secondary storage that uses lasers to store data.

solid-state storage: a type of secondary storage that uses transistors to store data.

REFLECTION

After completing Activity 3.7:

What kind of search data did you use to find out the data for the activity? Do you think you could have improved your search data?

In 1956, a technology company called IBM sold the first hard disk drive. It had a storage capacity of just 5 MiB of data and cost $50 000! Thankfully, hard disk drives have become much cheaper to buy and have a much larger storage capacity now. Another type of storage that became available to buy around 1982 is optical storage. People originally thought optical storage would replace hard disk drives, but soon realised that they could not be made with a high enough capacity to do this.

Optical storage uses lasers to read and write data to and from a circular disk. The disk spins and a read/write heads moves across the surface to allow data to be read or written. To write data, a laser burns pits into the surface of the disk. These pits are organised in a spiral track that starts at the centre of the disk. The surface of the disk between each pit is called a land. A laser is also used to read the pits and lands to obtain the data that they store. Examples of optical media are a CD, DVD and a Blu-ray. Two of those examples are read using a red laser and one is read using a blue laser, can you guess which is which?

ACTIVITY 3.8

Draw a diagram to represent the operation of optical storage.

Although optical storage is still used today, it is mostly used to archive data or as a cheap way to physically distribute files to users that require them. They didn't replace hard drives in the way that it was first thought they would. As portable devices developed, people began to realise there was a disadvantage with hard disk drives. This disadvantage came from the fact that they have moving parts. This made them vulnerable to wear and tear and also damage if they were dropped. As a result, a new type of storage was developed called solid-state storage.

Solid state storage doesn't have any moving parts. Instead it is created using **semiconductor chips**. Solid-state storage is sometimes known as flash memory. This is because of the ways that the data is read and written to the chips, it is said that it is 'flashed' onto them, but how does that happen? Solid state storage contains **cells** and **transistors** that are laid out in a grid. The transistors can be wired together in two different ways. In a NOR flash structure, the cells are wired together in a parallel structure. In a NAND flash structure, the cells are wired together in series. The intersection of the rows and columns of transistors are the cells. One of the transistors is called a **control gate** and one of the transistors is called a **floating gate**. The control gate sits on top of the floating gate. When a solid state storage device is first created, all the transistors are set to the binary value 1. The electric current reaches each control gate and then flows through to the floating gate to be stored. When data is stored in a transistor, it is converted to the binary value 0. A solid state drive is one example of a solid state storage device. You probably use another example on a regular basis too, it's called a USB flash memory drive (also known as USB stick, memory stick or a thumb drive).

ACTIVITY 3.9

You have learnt that a solid state drive and a USB flash memory drive are example of solid state storage. Can you think of another example?



Figure 3.11: CDs are a type of optical media

KEY WORD

semiconductor chips: an electronic circuit, where transistors are set on a semiconductor material, such as silicon.

COMPUTER SCIENCE IN CONTEXT

Organisations have a growing need to store more and more data. This data is often vital to the operation of the organisation. Data storage development companies are constantly trying to develop storage devices that have larger storage capacities. Due to this need, researchers have started to look at other methods of data storage including a revolutionary method called holographic data storage. This type of data storage would allow approximately a terabyte of data to be stored in a single crystal that is about the size of a grain of sugar.



Figure 3.12: A USB flash memory drive

KEY WORDS

cell: the intersection of the rows and columns of transistors in solid-state storage.

transistor: a device that acts as a switch or gate for electronic signals.

control gate: a component that controls the flow of electric current.

floating gate: a component that can store electrical charge.

virtual memory: a type of memory that can be used as an extension to the RAM.

page: a unit of data.

3.8 Virtual memory

So far, all the storage that you have learnt about has been physical storage. This means that it is an actual component storage unit that you can see and touch. There is another type of storage that can be created in a computer and this is called **virtual memory**. It is called virtual memory, as unlike physical storage, it isn't a component that you can see or touch.

You may have noticed that this chapter includes the term memory and storage. You may find these used interchangeably in lots of different resources. There is a slight difference between the two, but even the defined difference is a little contradictory.

Technically, memory is used as a term for short term temporary storage of data, and storage is used as a term for the permanent storage of data. For example, RAM stores data temporarily and is a type of memory, random access memory. A hard disk drive is a type of storage as it stores data permanently, until the user deletes it. This seems a simple distinction, but ROM is also a type of memory, read only memory, but it stores data permanently too. Therefore, the difference is not completely clear, but it is mostly considered that memory is temporary and storage is permanent. However, you don't need to become too worried about picking the correct term!

Virtual memory is also a type of memory that is created for temporary use. It is created as an extension to RAM. Sometimes, a computer is given a large number of tasks to complete at a given time. This means that it will need a lot of data to do this. All this data will need to be sent to the RAM to be ready to be processed. A computer system only has a set amount of RAM, so what happens when this RAM becomes full? It needs some extra help! This is where virtual memory can help.

Virtual memory is created in a computer system with a hard drive component. A section of that hard drive is partitioned off and is effectively an extra bit for RAM instead. When the RAM gets full, the computer system can transfer some data to the virtual memory to be held there until it is needed. It will choose to send data that isn't as immediately required as other data. The data that is transferred is collected in a structure called a **page**. These pages of data are transferred to the virtual memory to make sure there is enough room in the RAM for the immediate data that is being processed. Once the pages of data are required again, they can be sent back to the RAM to be processed. The diagram in Figure 3.13 shows the process:

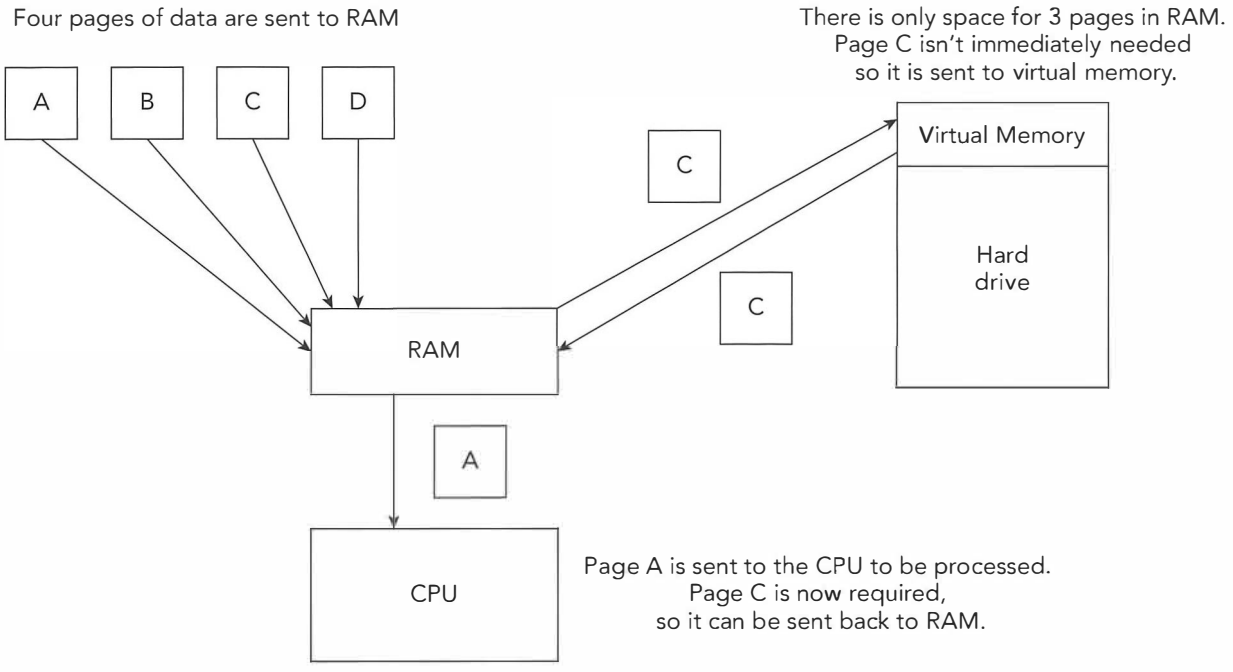


Figure 3.13: Pages of data are transferred to the virtual memory so that RAM doesn't become full

If the RAM became full and the computer did not have the ability to transfer pages of data to the virtual memory, this could cause it to crash, as it may not be able to complete the task it is currently working on.

3.9 Cloud storage

Despite a common misconception that many people have, **cloud storage** does not store our data somewhere up in the clouds. So how did it get that name? It is more in relation to the very large collections of data that are stored being like clouds of data, think clouds of millions of raindrops!

As technology has developed, and continues to do so, people have found that they want and need to store larger amounts of data, and data storage is quite expensive to buy, keep and maintain.

You might be able to make sure that your 2 Terabyte hard drive at home is safe in your house and is checked regularly for security issues, but it's likely you only have a relatively small amount of data to store. Some multinational businesses have huge amounts of data to store. This can require large rooms that are needed to house expensive hardware, such as computer **servers**, to store all the data that the company requires. This could be the details of millions of their customers and all the online transactions they have made to buy products. They would also need staff to maintain this equipment and keep that data safe. These kinds of costs could amount to many thousands of pounds a year for a business. Therefore, cloud storage offers them an alternative.

KEY WORDS

cloud storage: data storage that is owned by a third party and accessed by the user, using the internet.

server: a component that acts as a central resource for the storage of data and applications or for providing a service.

Cloud storage companies created a service in which they set up very large buildings that house and maintain thousands of computer servers. They spend 24 hours a day making sure the servers are maintained and secure. A company can then pay a fee to the cloud storage company for their service, to be allowed to store their data on their servers and pass the responsibility to the company to keep it secure until it is required. The company then accesses this data, when it is required, using the internet.

Cloud storage is a collection of third party hardware (hardware that is owned by another person or organisation), such as computer servers, that individuals and companies can use to store data, and for a range of other services. When data is stored within the buildings and hardware owned by the individual or business, this is known as storing the data locally. When data is stored in hardware owned by a third party and accessed using the internet, this is known as storing data remotely.



Figure 3.14: Data can be stored in cloud storage

Cloud storage began as a simple way to offload the responsibility of storing data. Since it was first used, it has developed into use for other services. For example, a business can now use cloud storage in connection to its website. The cloud storage company can offer a service whereby when demand is high for a website, such as they have a big sale, then they can use cloud storage to extend the capabilities of the web servers to be able to process the large amounts of data and requests that are being sent to the website. This means that the company can use hardware they do not own, on occasions when the demand for their business is high. This means they do not need to own all the hardware used, and have the extra hardware not being used, and therefore wasting money for them, when their business demand is lower. Helpful for business costs!

COMPUTER SCIENCE IN CONTEXT

Individuals are making increasing use of cloud storage. This allows portable devices to remain fairly small, as they do not need to store very large amounts of data locally within the portable device, e.g. their mobile phone. Companies such as Microsoft also offer access to software using cloud storage now, e.g. Office 365. This means that people do not need to download large software applications onto their portable devices.

Some people do have concerns about cloud storage. They think it is a big risk to give someone else the responsibility for looking after their data. Table 3.3 gives comparisons of some advantages and disadvantages of storing data in the cloud and storing it locally.

Storing data in the cloud	Storing data locally
You do not have to pay for the hardware to store the data.	You will need to purchase the hardware to store your data. These costs can be high if you have a lot of data to store.
You do not have the responsibility for the security of the hardware that stores the data. However, you are relying on a third party making sure that your data is safe. If they do not do this correctly, they may put your data at risk.	You will need to make sure that your data storage is well maintained and that it is regularly checked and updated for security issues.
You can access the data and resources using different devices from anywhere that you have an internet connection. However, if your connection fails or is not available, you will not be able to access your data. If your connection is slow, this can also mean that accessing your data is slow.	You do not need an internet connection to access your data, therefore it is available at all times that you have local access to that data.
You can increase and decrease the amount of storage space you need very easily, when you need to.	You might have redundant hardware, that you have paid for, that isn't used as the amount of data that you need to store has decreased.

Table 3.3: Comparisons of storing data in the cloud and locally

ACTIVITY 3.10

Do you store any data in cloud storage? If so, do you ever worry about doing this? Discuss with a partner about the advantages and disadvantages of storing data in the cloud, compared to local storage. Reach a conclusion about whether you think the risk is worth it.

As part of your discussion research at least one more advantage and one more disadvantage and make note of these. You can then share your thoughts from your discussion with the rest of your class and see who thinks cloud or local storage is better, and why.

3.10 Network hardware

At the beginning of this chapter, you learnt about the hardware required within a computer and local to a computer that enable the temporary and permanent storage of data. You have now learnt that data can be stored remotely, in a cloud storage system, but that an internet connection is required to access the data stored in the cloud. The hardware required for this connection creates a **network**. You will now learn what hardware is required to create a connection to other devices and the internet.

When two or more devices are connected, this is called a network. Some basic hardware devices are required to create a network, these are a **network interface card (NIC)**, a **router** and you may need some **transmission media**.

KEY WORDS

network: computers and devices that are joined together using cables or wireless technology.

network interface card (NIC): the component in a computer that is used to connect it to a network.

router: a network component that examines a data packet to obtain its destination address and then forwards the packet to this address.

transmission media: the cables that are used to transfer data in a network.

Within your computer or device is a piece of hardware called a NIC. This may be a wired or wireless NIC. This means that either a cable will be plugged into the hardware to connect to a network, or it will have a wireless component that allows it to make a wireless connection. Most modern NICs are wireless NICs (WNICs).

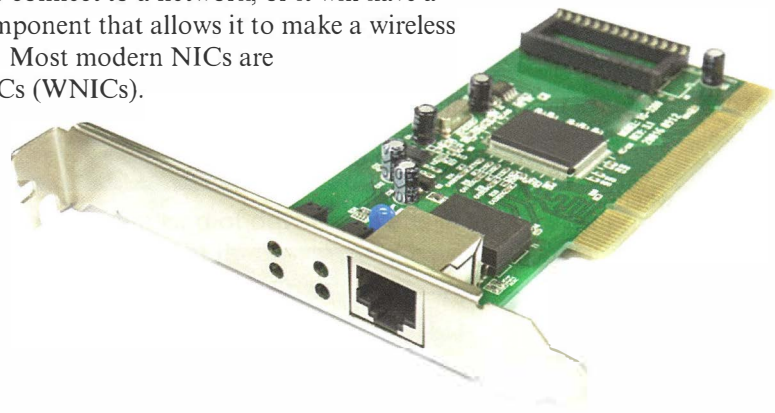


Figure 3.15: A network interface card (NIC)

Each NIC is given an address called a **media access control (MAC) address**. This address can be used to identify a device on a network.

A MAC address is assigned by the manufacturer and is never normally changed. It will remain assigned to the device whilst it is in use. A MAC address is made up of two main parts. These are the manufacturers ID code and the Serial ID code for the device. This creates a unique address for each device. The address is represented as hexadecimal and normally has six pairs of numbers, separated with either a dash or a colon, for example:

34:4D:EA:89:75:B2

When a device with a network interface card transmits data across a network, it will send the MAC address of that device with the data. This means that it is possible to tell, from inspecting the data, which device on the network the data was sent from.

KEY WORD

media access control (MAC) address: the unique address that is given to a NIC by the manufacturer.

ACTIVITY 3.11

See if you can find out the MAC address of one of your devices, such as a computer, laptop or a mobile phone.

If you do not know how to find it, you can use the internet to find a tutorial.

When data is sent across a network, a hardware device can be used to make sure that the data arrives at the correct destination. You can use a hardware device called a router to do this. Will a router have a NIC? Does this mean it will have a MAC address? See if you can find out! Each device that wants to connect to the internet will also need to do this by connecting to the router. This could be a wired or wireless connection.

A router is a hardware device that reads each **data packet** that it receives. Details about data packets and their structure was covered in Chapter 2. The router reads the data packets to find out the destination of the data. It can use the MAC address to do this,

KEY WORD

data packet: a unit of data that is used to transmit data across a network.

but that will normally only be used if the data is sent to another device that is part of the same local network (this is a network that is contained within a specific area, such as your house). As a router is also used to connect a device or a local network to the internet, it is the internet protocol (IP) address of the destination device that is normally read and used to forward the data to its correct destination.

This IP address can also be used to locate your device on a network, and it is also a unique address. It is different in structure to a MAC address, as it is assigned by the network, rather than by the manufacturer. In a local network, such as a home network, it is often the router that assigns the IP address to a device, when it is connected to the router.



Figure 3.16: A home router

An IP address can have different structures, such an IPv4 or IPv6 structure. IPv4 was the first structure created for an IP address. This structure was created back in 1983 and uses 32-bits to create each numerical IP address. Even though it is now an old structure, there are still many devices that have an IPv4 address. An example of an IPv4 address is:

12.244.233.165

As more and more devices started connecting to networks, the number of available IPv4 addresses started to rapidly decline. As each IP address needs to be unique to the device, they were rapidly running out, due to the number of devices worldwide. IPv6 was created as result, to create more available IP addresses. IPv6 uses 128-bits to create an IP address. IPv6 addresses can also contain hexadecimal characters. This created much more possibilities for IP addresses and allowed them to remain unique to the device. An example of an IPv6 address is:

2001:0db8:0000:0000:0000:ff00:0042:7879

You will also notice that an IPv4 address is separated by full stops, but an IPv6 address is separated by colons.

An IP address can also be either **static** or **dynamic.** Most devices are assigned a dynamic IP address. This means that the address can change each time the device is connected to a network. The device may be given that IP address for a single session

KEY WORDS

static IP: an IP address that does not change each time the device connects to a network.

dynamic IP: an IP address that can change each time the device connects to a network.

that it connects to the router, or it may be given it for several sessions before it changes, but it will change at some point. It is possible to obtain a static IP address for your device. This means that the IP address for your device will be the same every time that you connect your device to a network, it will not change, until you request it to be changed, it will always remain the same. It is normally your Internet Service Provider that will provide you with a static IP address for your device.

ACTIVITY 3.12

Can you find out what kind of devices a static IP address is normally used for? Can you also find out what the benefit is of having a static IP address?

SUMMARY

Hardware is all the physical components that you can see and touch in a computer system.
The CPU is the component in a computer system that processes all the instructions and data. It uses an instruction set to do this.
The CPU has several component parts, these include the ALU, CU, PC, MAR, MDR, CIR, ACC and buses.
The CPU is responsible for carrying out the fetch-decode-execute cycle.
The number of cores, the clock speed and the cache size can all affect the performance of the CPU.
Computers also need input and output devices to be able to provide a system with data, then obtain the results when it has been processed.
Sensors are a type of input device that can be used to capture data from the immediate surrounding environment.
A computer has both primary and secondary storage.
Primary storage is directly accessed by the CPU and consist of RAM and ROM.
Secondary storage is not directly accessed by the CPU and there are three types, magnetic, optical and solid-state.
Virtual memory is created by portioning part of the hard drive. It is used when RAM is full.
Cloud storage is when data is stored remotely on servers and storage that is owned and maintained by a third party.
A network is created when two or more devices are connected.
A device needs a NIC to connect to a network. Each NIC has a MAC address.
A router is used in a network to forward packets of data to their correct destination. A router can assign an IP address to each device.
There are two types of IP address, IPv4 and IPv6.

EXAM-STYLE QUESTIONS

1 A student has a computer that has a Von Neumann architecture.
The CPU in the computer performs the fetch-decode-execute cycle to process data and instructions.

a i **Identify** the name of the component where an instruction is stored immediately before it is fetched into the CPU. [1]

ii Identify the name of the component that stores the address of the next instruction that will be processed. [1]

iii **Describe** what happens at the decode stage of the cycle. [3]

b The computer has a 2 GHz single core processor. The students wants to improve the performance of the computer by replacing the CPU.
The student selects a new CPU that is a 3.4 GHz dual core processor.

i Identify how many cores are in the new CPU. [1]

ii **Explain** why the performance of the new CPU would be better. [2]

c The computer has both random access memory (RAM) and read only memory (ROM).

i Identify if these are examples of primary or secondary storage. [1]

ii **State** what each of these components store. [2]

[Total: 11]

2 **Give** the most appropriate sensor for use in each of the given scenarios. A different sensor must be given for each scenario. [3]

Scenario	Most appropriate sensor
To monitor the amount of pollution in a river.	
To count the number of competitors who finish a race.	
To detect a person who is approaching an automatic door system.	

3 A small business uses optical data storage to archive their customers' data.

a i Give *two* examples of optical storage. [2]

ii Describe how data is stored on optical storage. [6]

b The business currently has a local archive for their data. They are considering storing their customer data in cloud storage instead.

i Give *one* advantage of storing the customer data in cloud storage, rather than local storage. [1]

ii Give *two* disadvantages of storing the customer data in cloud storage, rather than local storage. [2]

[Total: 11]

4 A computer connects to the internet and is assigned an IP address.
The IP address has an IPv4 structure.
Describe the structure of an IPv4 address. [2]

COMMAND WORDS

identify: name / select / recognise

describe: state the points of a topic / give characteristics and main features.

explain: set out purposes or reasons / make the relationships between things evident / provide why and/or how and support with relevant evidence.

state: express in clear terms.

give: produce an answer from a given source or recall / memory.

SELF-EVALUATION CHECKLIST

After studying this chapter, think about how confident you are with the different topics.
This will help you to see any gaps in your knowledge and help you to learn more effectively.

You might find it helpful to rate how confident you are for each of these statements when you are revising.
You should revisit any topics that you rated 'Needs more work' or 'Getting there'.

I can...	See topic	Needs more work	Getting there	Confident to move on
explain why a computer system needs a CPU.	3.1			
describe the role of each component in a CPU.	3.1			
explain how data is fetched into the CPU, including the components used.	3.2			
explain how data is decoded and executed by the CPU, including the components used.	3.3			
explain what an embedded system is and identify examples of where one might be used.	3.1			
describe the purpose of an input device and provide examples.	3.6			
describe the purpose of an output device and provide examples.	3.6			
identify the different types of data that is captured by different sensors.	3.6			
choose a suitable sensor for a given scenario.	3.6			
give characteristics of primary data and explain the role of RAM and ROM.	3.7			
give characteristics of secondary storage.	3.7			
describe how data is written to and read from magnetic storage.	3.7			
describe how data is written to and read from optical storage.	3.7			
describe how data is written to and read from solid state storage.	3.7			
explain how virtual memory is created and when it is needed in a computer system.	3.8			
describe what cloud storage is and how it is used.	3.9			
explain the advantages and disadvantages of storing data in the cloud, rather than storing it locally.	3.9			